Analogue electronics revision summary

National 5 · Electronics & control · Answers

Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1 Name the three connections of an NPN transistor, and state the voltage needed at the base to switch it on.

- Base, collector and emitter
- About 0.7 V

2 Explain why a relay is used in a control circuit.

- It lets a small current from the control circuit switch a separate, higher-current or higher-voltage circuit

3 State what happens to the resistance of an LDR as it gets darker, and of a thermistor as it gets hotter.

- LDR: resistance rises as it gets darker
- Thermistor: resistance falls as it gets hotter

4 A 4 kΩ resistor and a 6 kΩ resistor form a divider on a 10 V supply. Calculate V_{out} across the 6 kΩ resistor.

- $V_{out} = 6 \div 10 \times 10 = 6$ V

5 Complete the table: draw each symbol, say whether it is an input or output transducer (or neither), and what it does.

- Battery: neither; supplies electrical energy
- Switch: input; opens or closes the circuit
- Resistor: neither; limits current
- Variable resistor: input; adjustable resistance, e.g. to set sensitivity
- LDR: input; resistance falls as light increases
- Thermistor: input; resistance falls as temperature rises
- Diode: neither; lets current flow one way only
- LED: output; gives light
- Lamp: output; gives light
- Buzzer: output; gives sound
- Motor: output; gives movement